

Session - 01 // Objective

✓ ☒ Why maths

☐ Can we solve business problem using only maths?

☒ Maths $\approx$ ML/AI?

☒ Linear Alg? $\rightarrow$ Basic $\rightarrow$ Use in AI/ML

☒ Implementation ✓

☒ Learn to Read Research paper: $\rightarrow$ Implementation

Fine tuning $\rightarrow$

90/50

data

data points

label

cid

✓

✓

✓

✓

✓

5

6

7

income

(17

(6

(12

(13

(17

(3

4

city score

(50)

(45)

(50)

(60)

(44)

(90)

469

prob

status

Yes

No

Yes

Yes

Yes

No

No

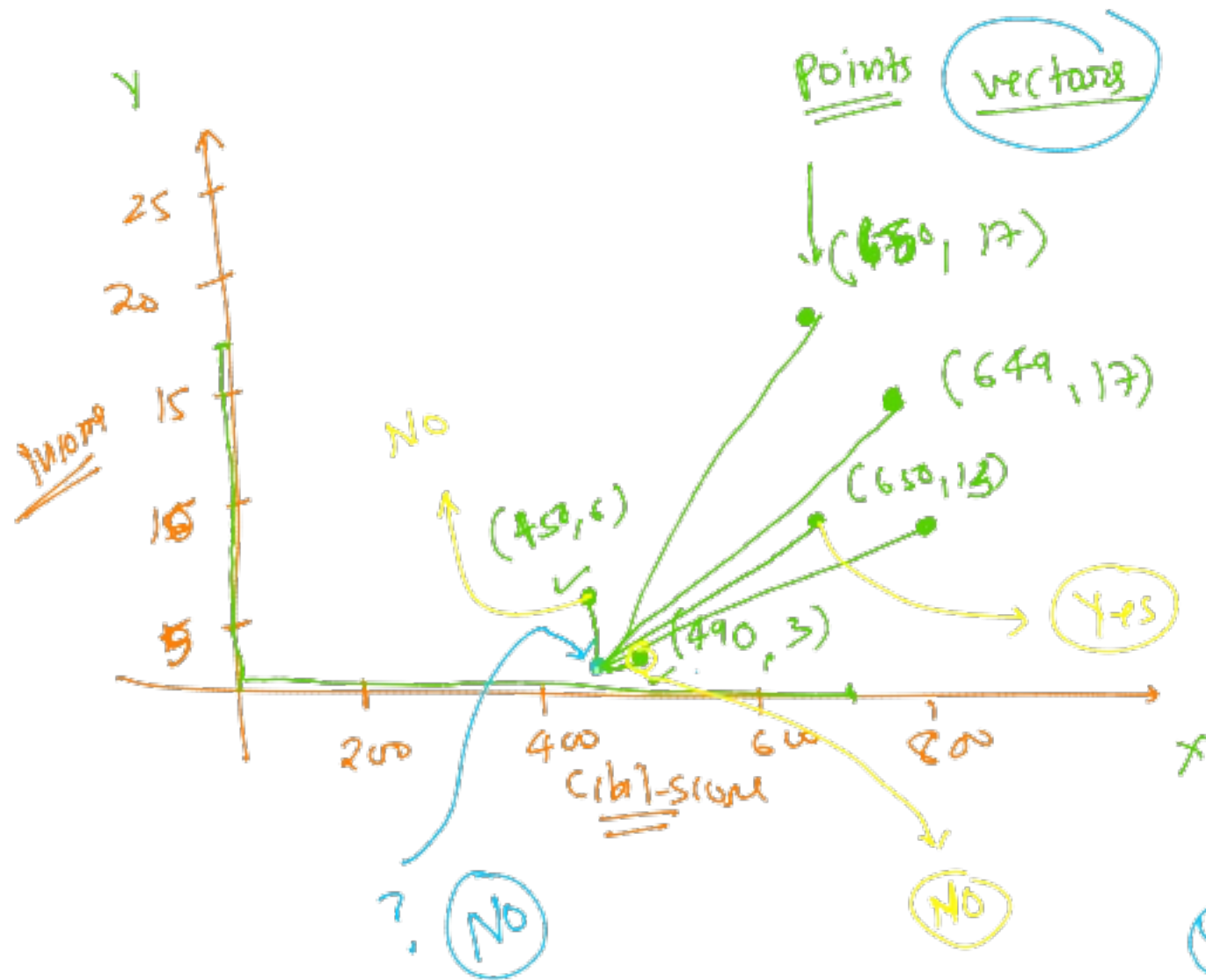
?

Yes or No

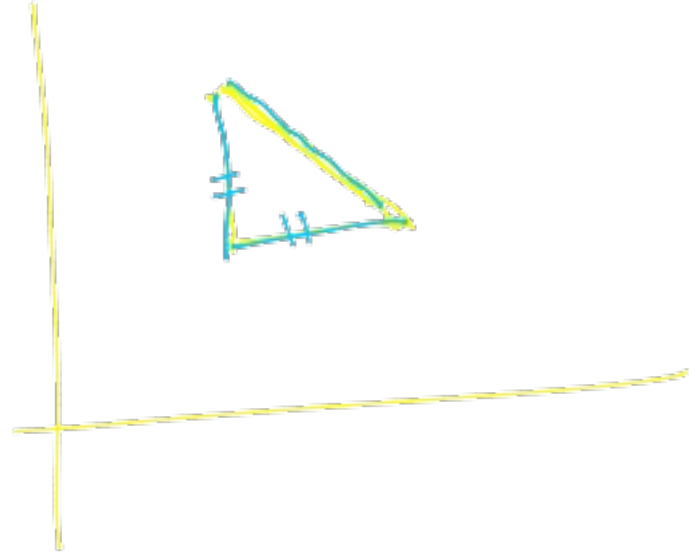
$C_P(17, 880)$

$$C_1(650, A)$$

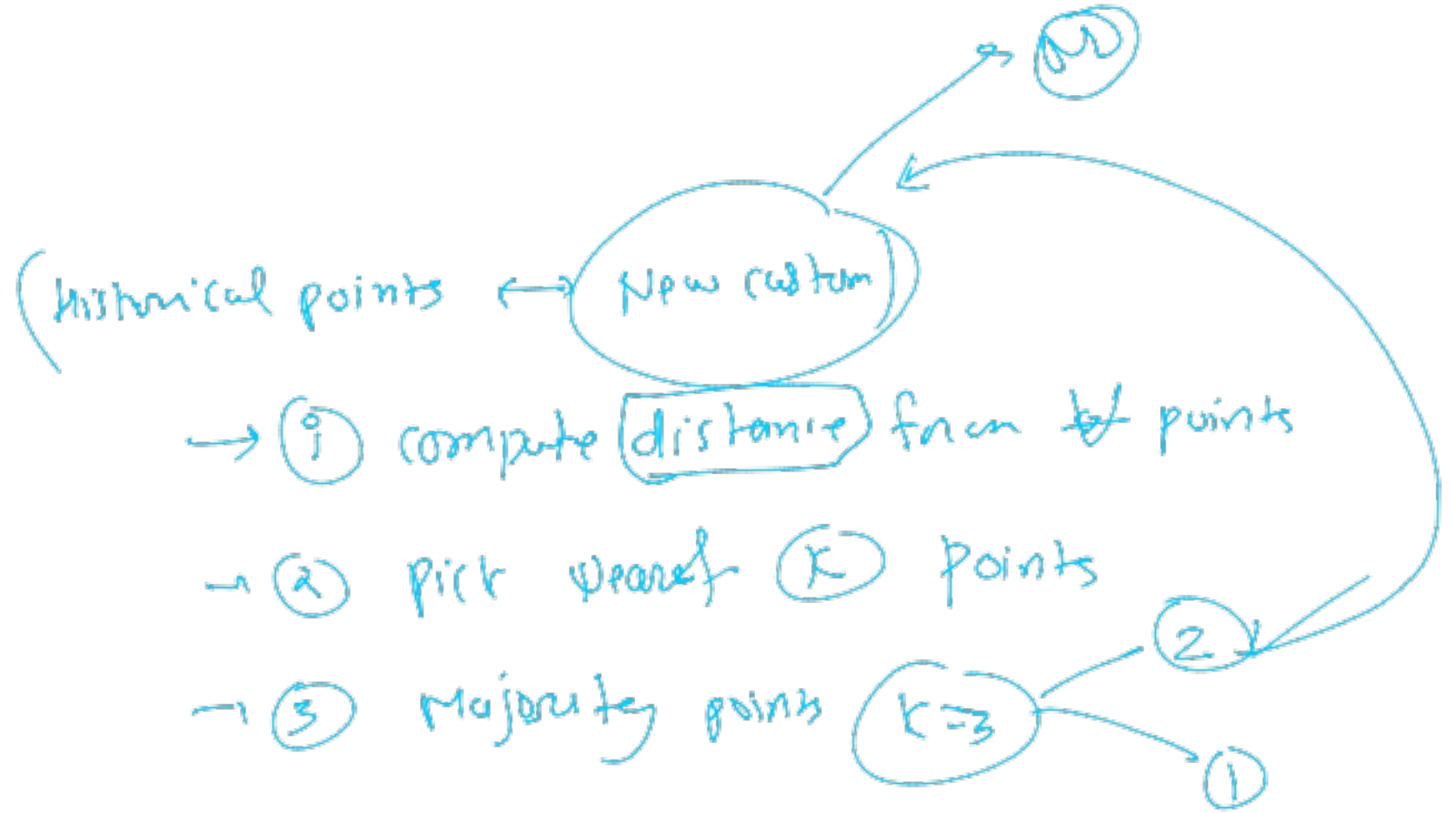
[4]

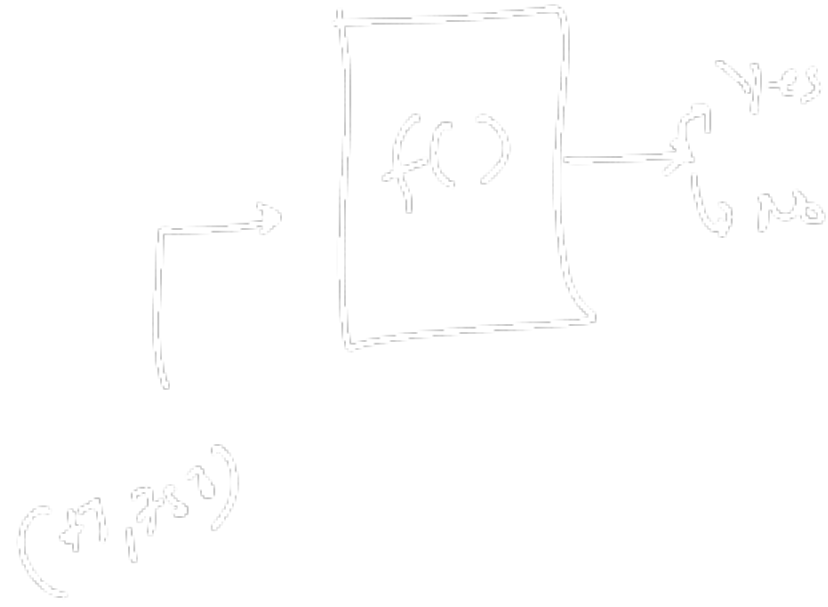

$$\begin{cases} \text{No} \rightarrow 2 \\ \text{Yes} \rightarrow 1 \end{cases}$$
$$\frac{2}{3} \rightarrow \frac{6}{9}$$


$\frac{1}{3} \Rightarrow 37.3$



python fn





Distance b/w points / vector from origin



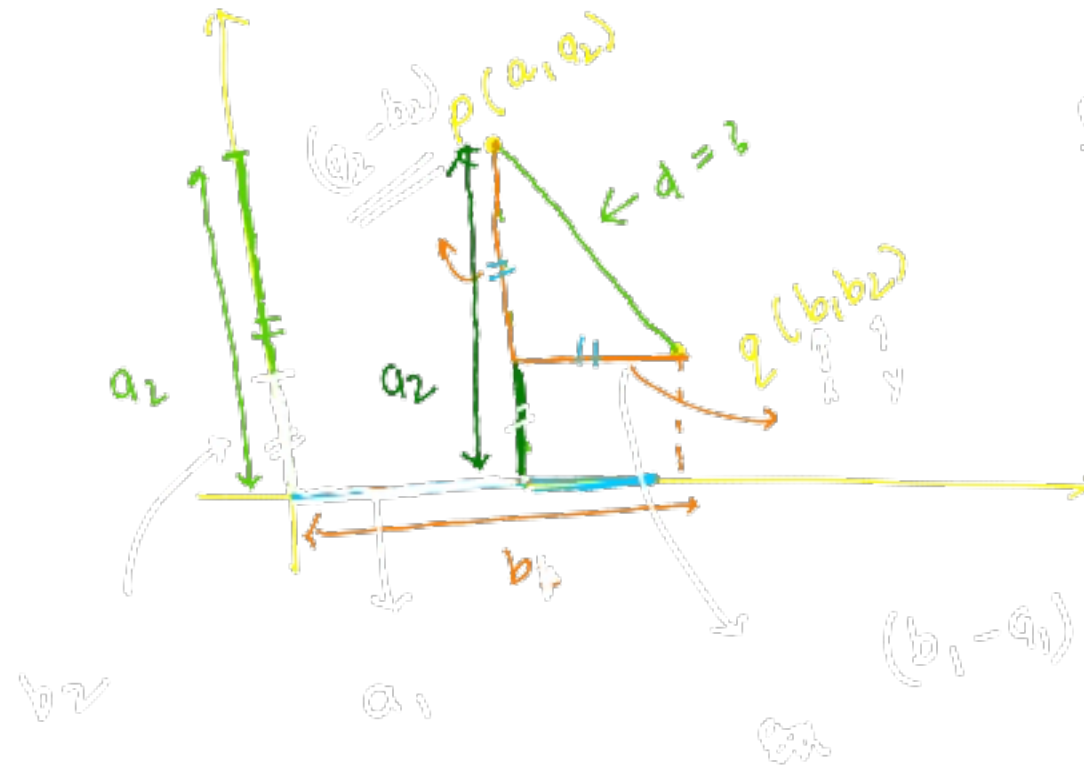
$$d^2 = a^2 + b^2$$

$$d = \sqrt{a^2 + b^2}$$

Euclidean distance

Distance b/w ② points

②D



$$d = \sqrt{(a_2 - b_2)^2 + (a_1 - b_1)^2}$$

$$\begin{matrix} P = (a_1, a_2) \\ \updownarrow \\ Q = (b_1, b_2) \end{matrix}$$

$$d = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2}$$

3D

$$p := (a_1 \ a_2 \ a_3)$$

$$q := (b_1 \ b_2 \ b_3)$$

$$d = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2}$$

any dim

→

$$d = \text{sqrt} \left(\sum_{i=1}^n (a_i - b_i)^2 \right)$$

nD

--- -- -- -- --

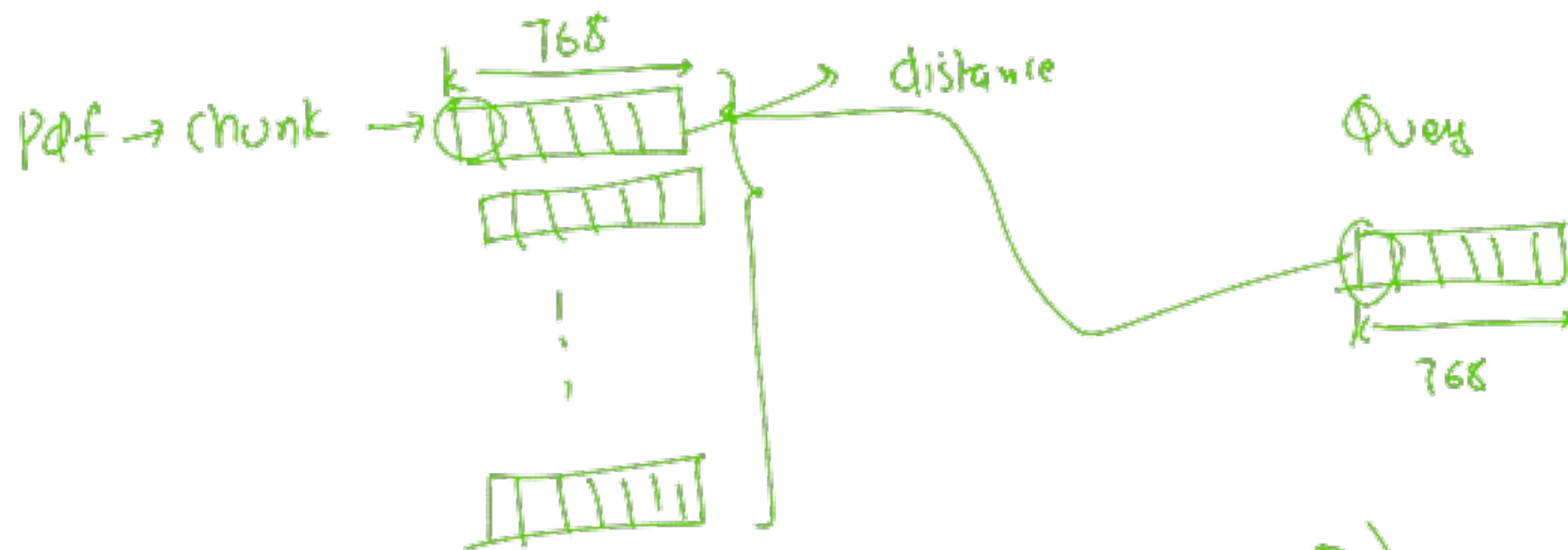
$$p := (a_1 \ a_2 \ a_3 \ \dots \ a_n)$$

$$q := (b_1 \ b_2 \ b_3 \ \dots \ b_n)$$

↓↓

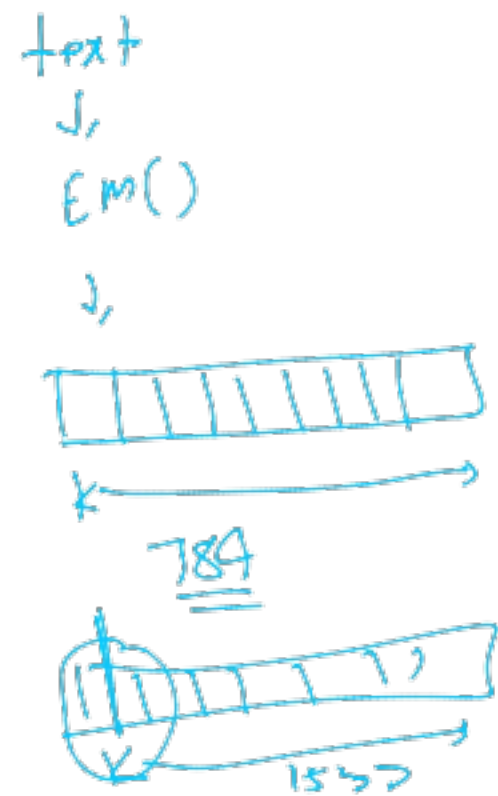
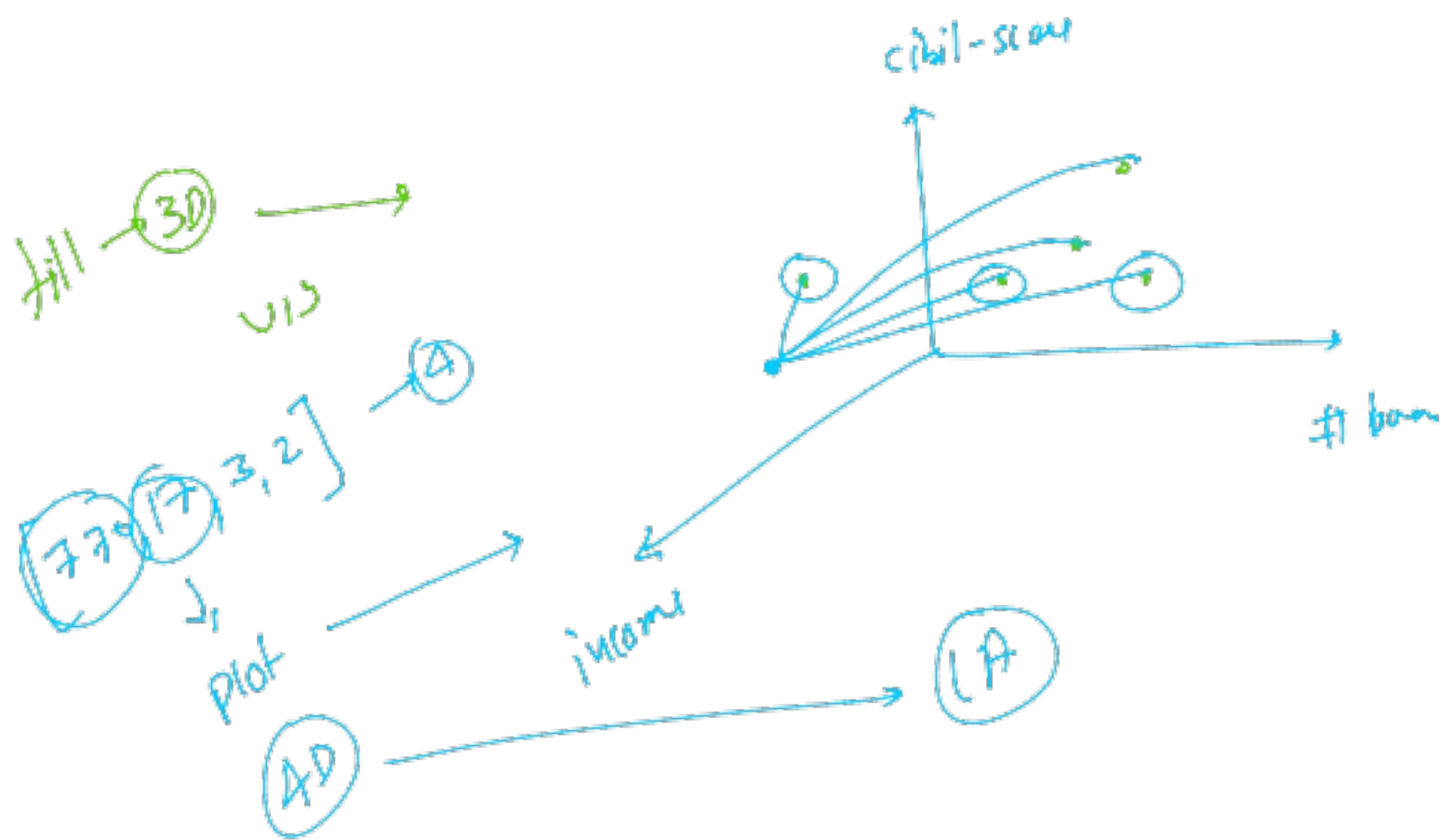
$$d = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + \dots + (a_n - b_n)^2}$$

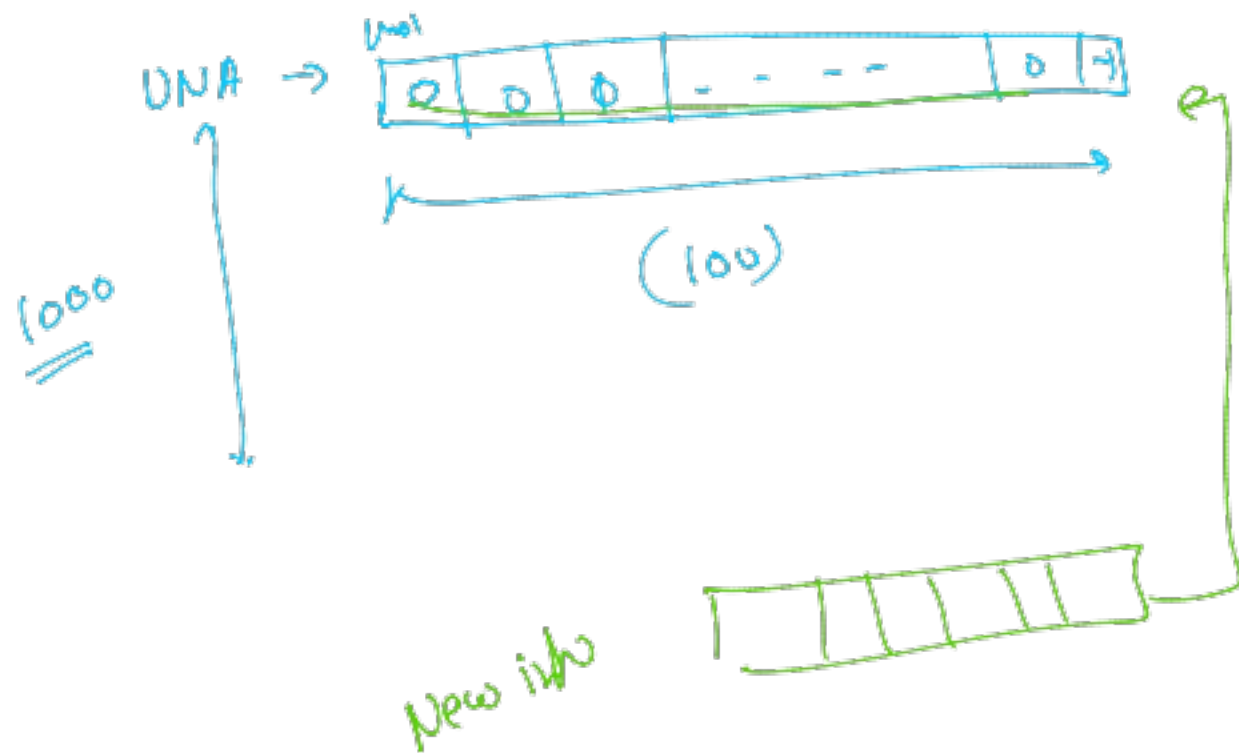
↓



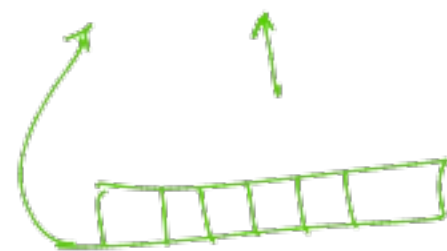
$$d = \sqrt{\sum_{i=1}^n (q_i - b_i)^2}$$

cid	credit score	income	# loan	status	# closed loan	sm
	(170	17	3)	yes		

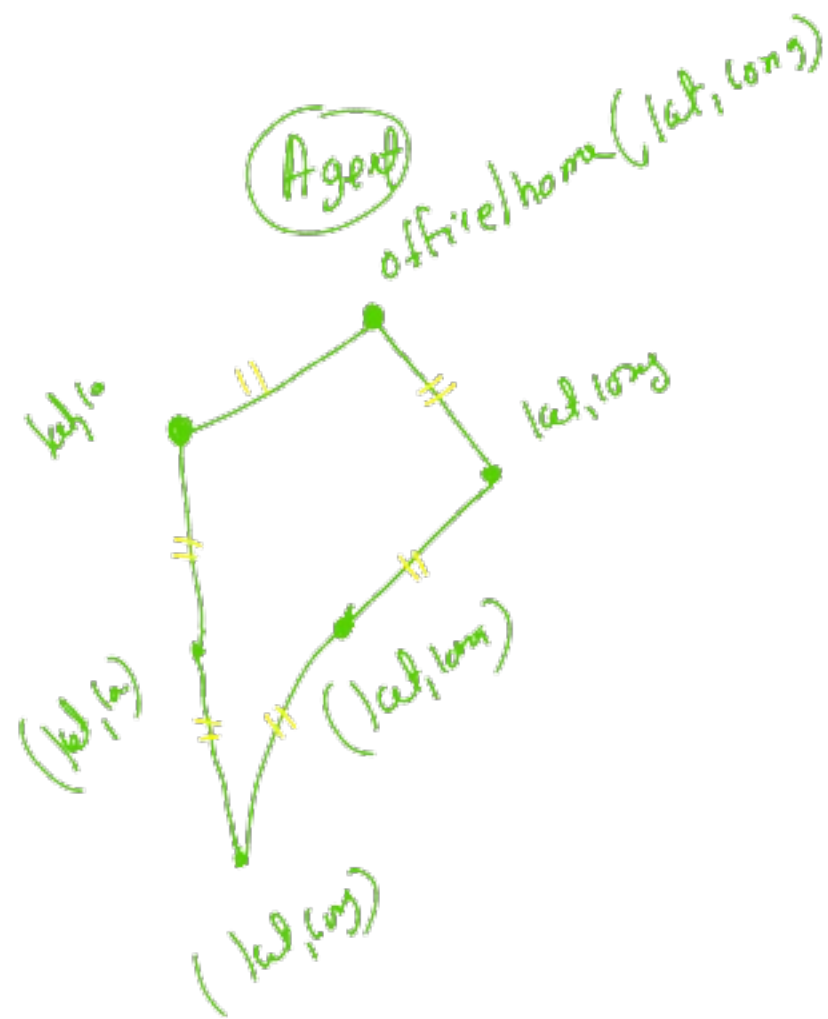




(i) Hamming distance



(ii) Haversine distance



total distance

8

Vector / points

↓

(i) add

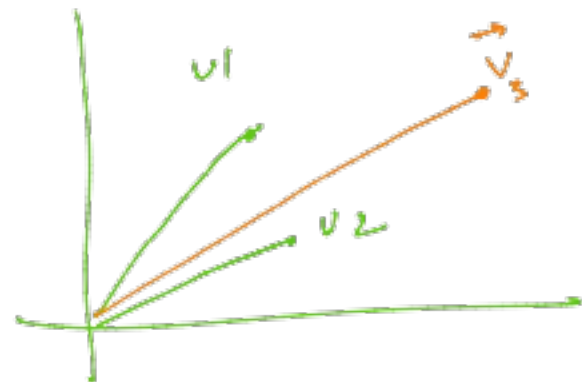
(ii) sub

$$\Rightarrow [17, 15]$$
$$v_2 \Rightarrow [16, 15]$$
$$v_3 \Rightarrow [33, 30]$$

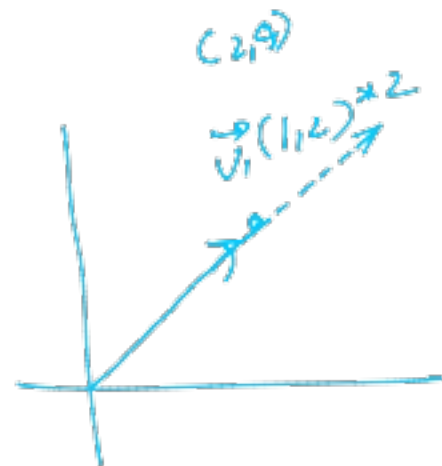
dot product

~~Cross product~~

* $\chi(w)$ mol



manim



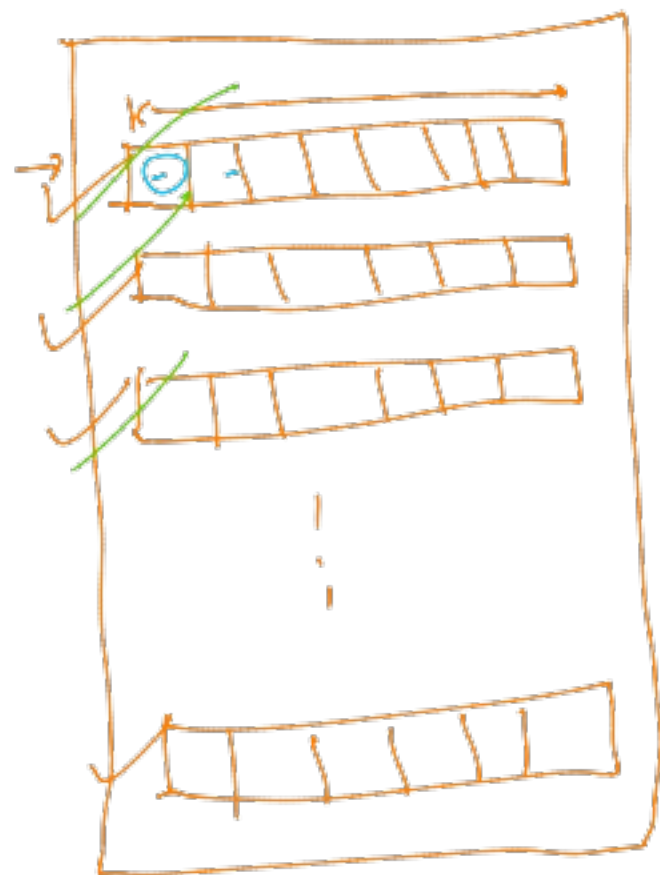
pdf

chunk $\rightarrow$ EM()

(100)



ANN



vector store

Question



~~$d(q, c) \rightarrow 0.08$~~

~~$\sqrt{(c - q)^2 + () + () \dots}$~~

~~for loop~~

Quash
↓



Application
(SaaS)

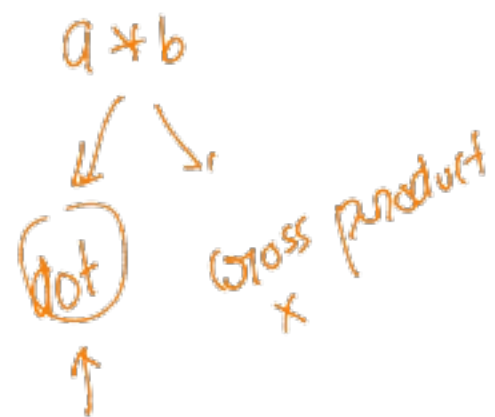


bLM



manim





$$a = [a_1, a_2, a_3]_{1 \times 3}$$

$$b = [b_1, b_2, b_3]_{1 \times 3}$$

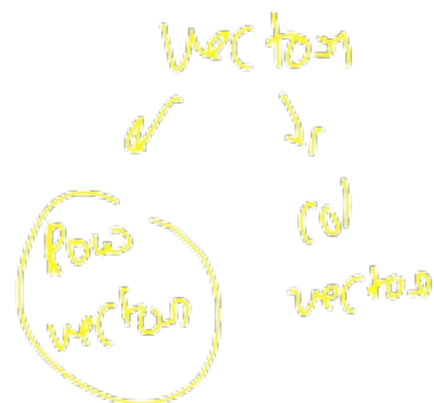
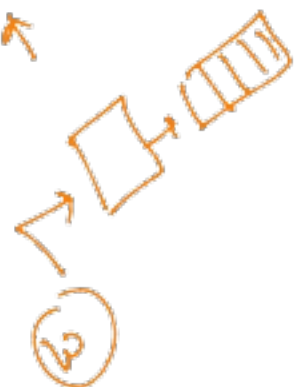
$$w_1 = [2, 3, 1]$$

$$w_2 = [2, 3, 0.5]$$

matrix

$$\begin{bmatrix} [1, 2, 3] \end{bmatrix}$$

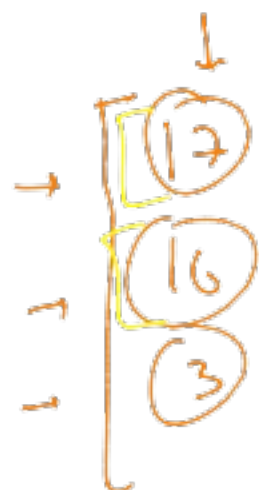
word vec



$[[7, 650] \checkmark$

$[16, 750] \checkmark$

$[3, 350] \checkmark$



order

list of list

$$(A)_{1 \times 3}$$

$$(B)_{1 \times 3}$$

$$(w_1, w_2) \Rightarrow \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 6 & 2 & 1 \end{bmatrix}$$

(3D) ↓↓

$$2 \times 6 + 3 \times 2 + 1 \times 1$$

=

↓

$$12 + 6 + 1$$

$$\downarrow$$

$$\textcircled{19}$$

eg.

$$w_1 \Rightarrow [2, 3, 1]_{1 \times 3}$$

$$w_2 = [6, 2, 1]_{1 \times 3}$$

$$\textcircled{A \times B}$$

$$(A)_{m \times n}$$

$$(B)_{n \times p}$$



$$a \cdot b = a^T \cdot b$$

↗
11,

$$\sum_{i=1}^n a_i \cdot b_i$$

$$a = (a_1 \ a_2 \ a_3 \ \dots \ a_n)$$

1 4

$$b = (b_1 \ b_2 \ b_3 \ \dots \ b_n)$$

$$a^T \cdot b = \sum_{i=1}^n a_i \cdot b_i$$

↑

Geometrically

Meaning

$a(a, b)$

$b(b, c)$



$$\underline{\underline{cost}} \Rightarrow [-69, 10]$$

$$(apple \cdot mango) \Rightarrow 9 \times 10 + 6 \times 7$$

$$\Rightarrow 90 + 42 \Rightarrow 132 \uparrow$$

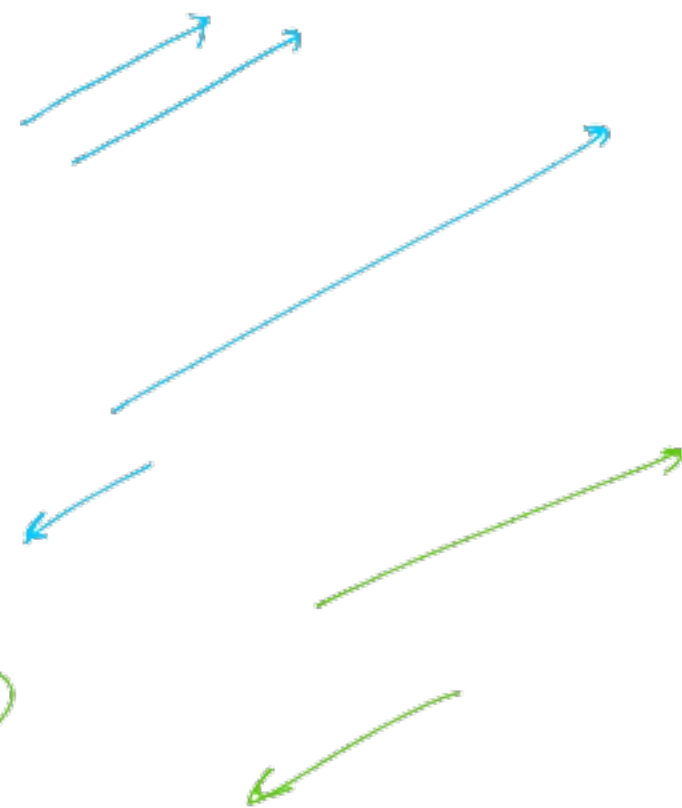
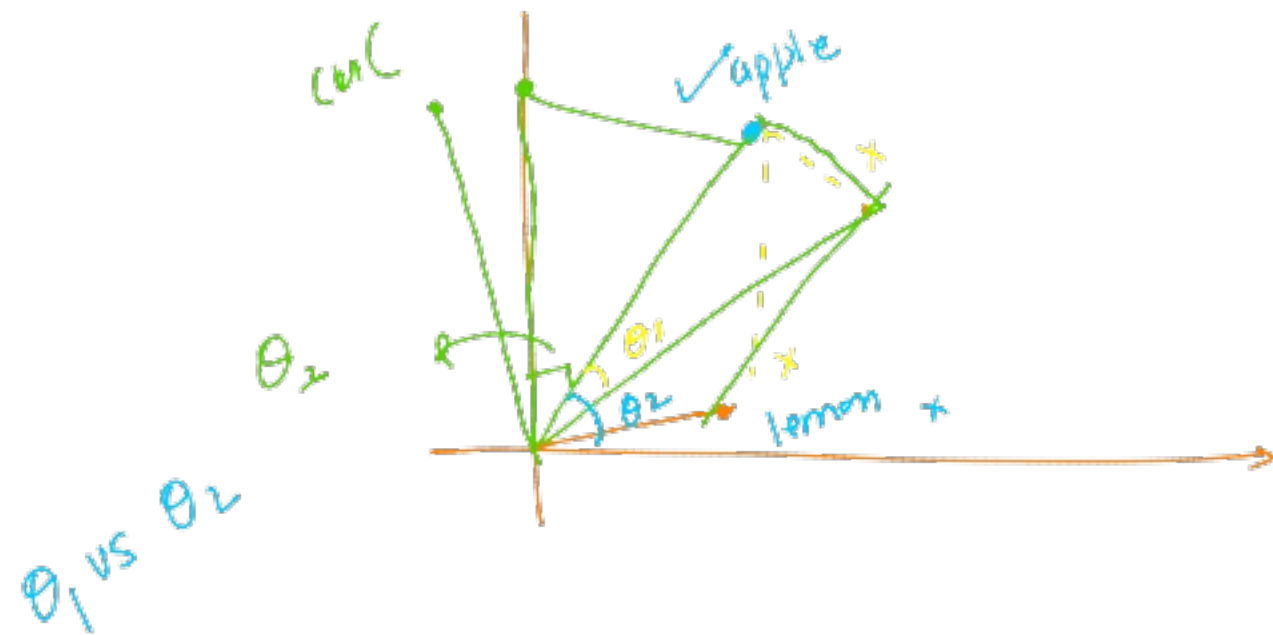
$$\underline{\underline{(apple \cdot banana)}} = 9 \times 2 + 6 \times 2 \Rightarrow 18 + 12 \Rightarrow 30$$

~~cost~~ $\Rightarrow$
~~ca~~
~~avg~~

$(0, -10)$
 $(6, -10)$

✓ ~~apple~~ $\rightarrow [9, 6]$
✓ ~~mango~~ $\rightarrow [10, 7]$

✓ ~~banana~~ $\rightarrow [2, 2]$

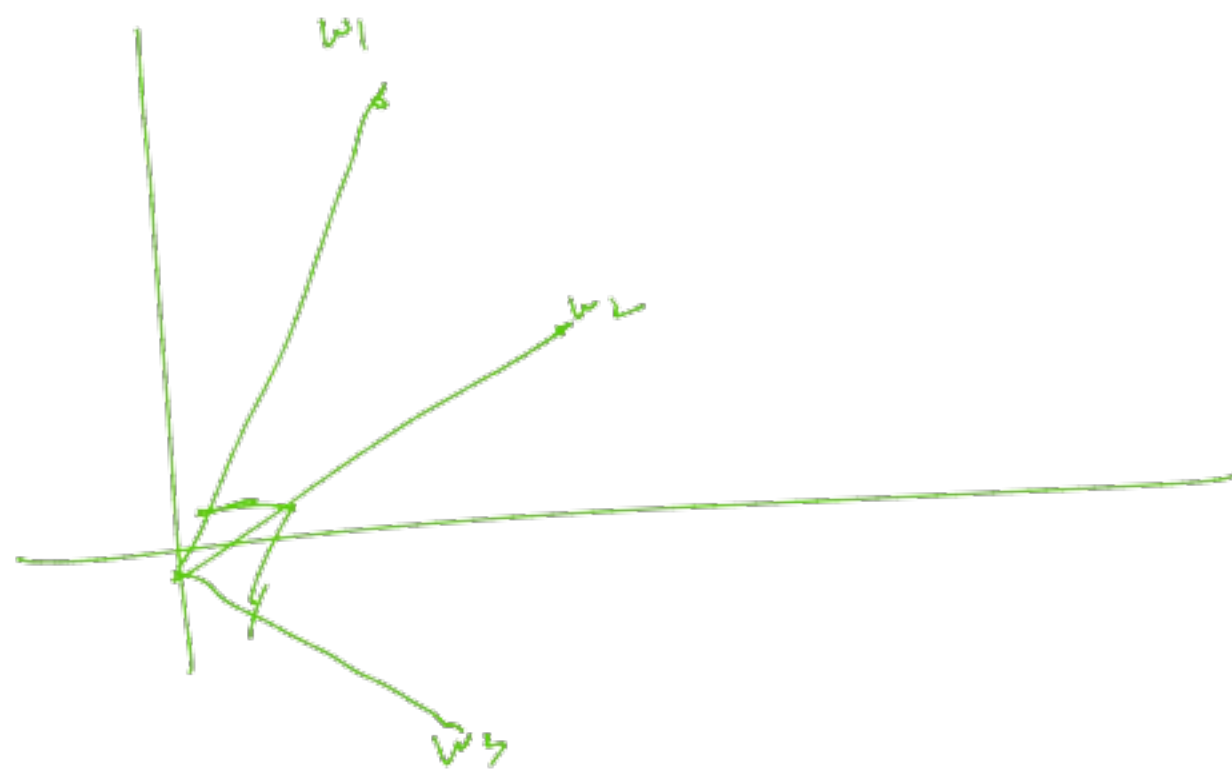


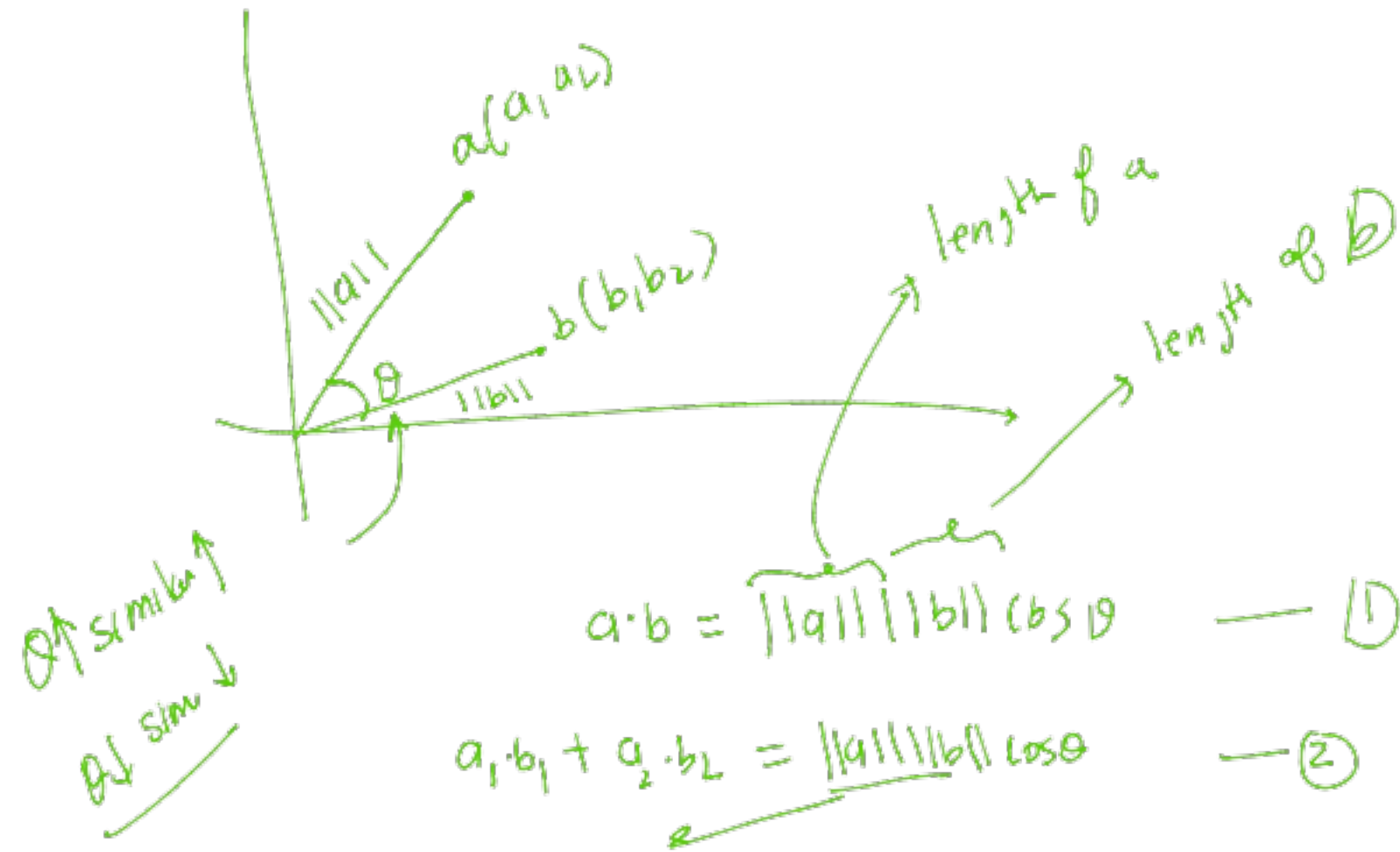
$$\begin{matrix} \text{apple} \cdot \text{car} \Rightarrow & 9 \times 0 + 10 \times 6 \Rightarrow & \textcircled{60} \\ \begin{pmatrix} 9, 6 \end{pmatrix} & \begin{pmatrix} 0, 10 \end{pmatrix} & \end{matrix}$$

main | vector

$$w_1 \Rightarrow [a_1 \ a_2 \ a_3]$$

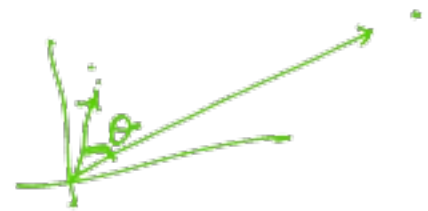
$$w_1^T = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}_{3 \times 1}$$

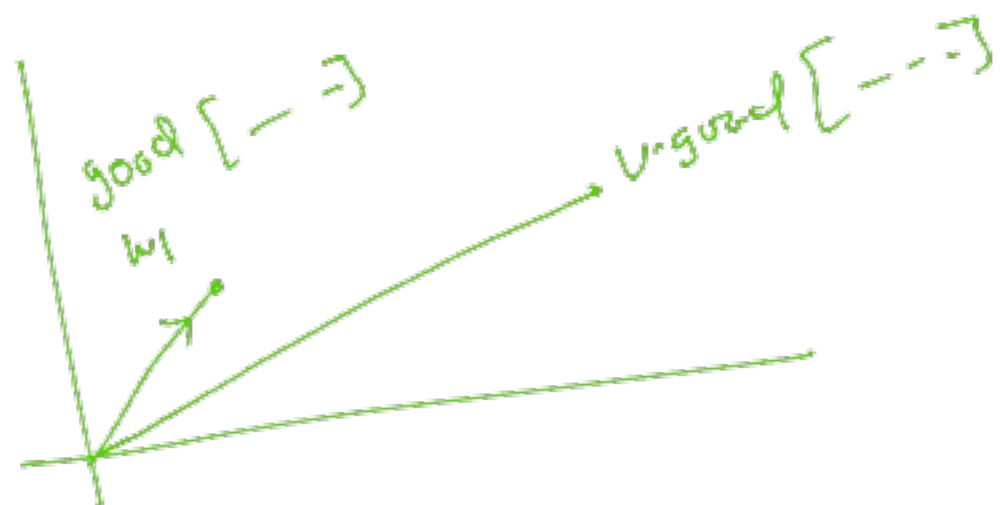


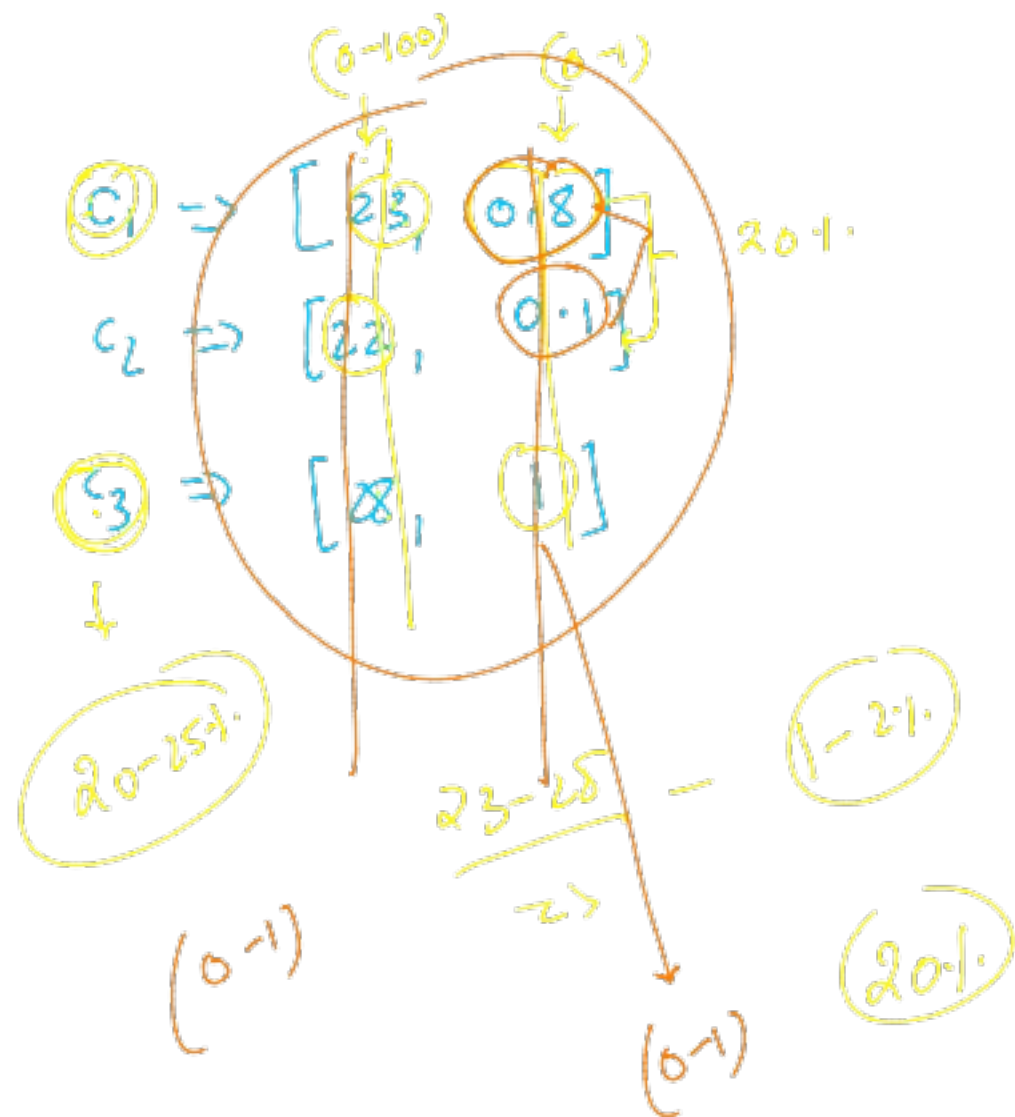
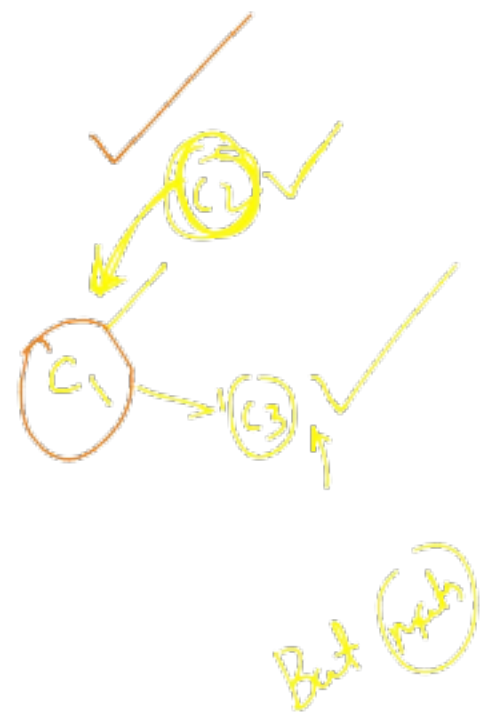


$$\cos \theta = \frac{a_1 \cdot b_1 + a_2 \cdot b_2}{\|a\| \|b\|}$$

$$\theta = \cos^{-1} \left(\frac{a_1 \cdot b_1 + a_2 \cdot b_2}{\|a\| \cdot \|b\|} \right)$$

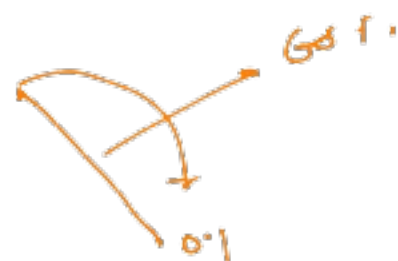


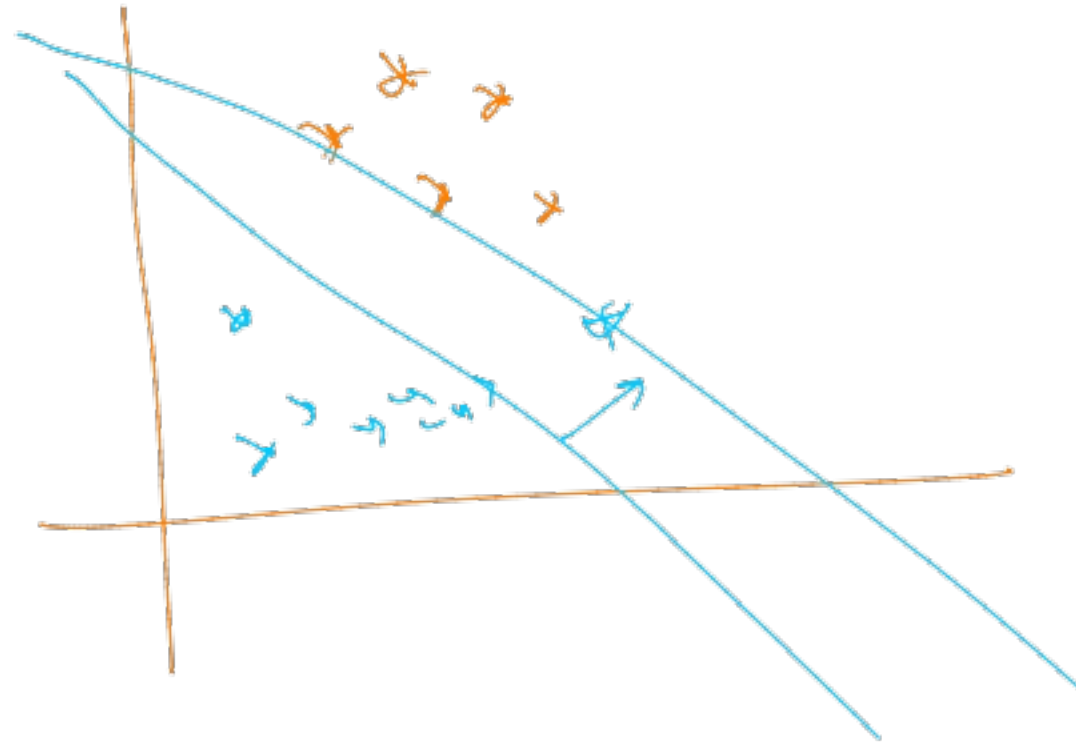




$$\frac{0.8 - 0.1}{0.8}$$

60-70.1





w2vec model
 (2013)
 (wikipedia page)
 1
 5 topic



6

I	am	good
↓	↓	↓
w _I	w _L	w _s
↓	↓	
[0.0 0.0]	[-1]	

last

New

